

Program:

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#include<stdio.h>
#define MAX_SIZE 100
void createArray(int arr[], int *size);
void insertElement(int arr[], int *size, int element, int position);
int searchElement(int arr[], int size, int element);
void deleteElement(int arr[], int *size, int position);
void displayArray(int arr[], int size);
int main() {
int arr[MAX_SIZE];
int size = 0;
int choice, element, position, result;
while(1) {
printf("\nArray Operation Menu:\n\n");
printf("—1. Create Array\n\n");
printf("—2. Insert Element\n\n");
printf("—3. Search Element\n\n");
printf("—4. Delete Element\n\n");
printf("—5. Display Array\n\n");
printf("—6. Exit\n\n");
printf("—Enter your choice: —");
scanf("—%d\n", &choice);
switch(choice) {
case 1:
createArray(arr, &size);
break;
case 2:
printf("—Enter element to insert: \n");
scanf("—%d\n", &element);
printf("—Enter position to insert (0 to %d): \n", size);
scanf("—%d\n", &position);
insertElement(arr, &size, element, position);
break;
case 3:
printf("—Enter element to search: —");
scanf("—%d\n", &element);
result = searchElement(arr, size, element);
if(result != -1)
printf("—Element found at position: %d\n\n", result);
else
printf("—Element not found in the array\n\n");
break;
case 4:
printf("—Enter position to delete (0 to %d): \n", size-1);
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scanf("%d", &position);
deleteElement(arr, &size, position);
break;
case 5:
displayArray(arr, size);
break;
case 6:
return 0;
default:
printf("Invalid choice! Please enter a valid option.\n");
}
}
return 0;
}

void createArray(int arr[], int *size) {
int n, i;
printf("Enter the number of elements: ");
scanf("%d", &n);
if(n > MAX_SIZE) {
printf("Error: Number of elements exceeds the maximum array size.\n");
return;
}
printf("Enter %d elements: ", n);
for(i=0; i < n; i++) {
scanf("%d", &arr[i]);
*size = n;
printf("Array created successfully.\n");
}
}

void insertElement(int arr[], int *size, int element, int position) {
int i;
if(*size >= MAX_SIZE) {
printf("Error: Array is full. Cannot insert element.\n");
return;
}
if((position < 0) || (position > *size)) {
printf("Error: Invalid position.\n");
return;
}
for(i=0; i < *size; i++) {
arr[i] = arr[i + 1];
}
arr[position] = element;
(*size)++;
}

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printf(—Element inserted successfully.\n\n);
}
int searchElement(int arr[], int *size, int position) {
int i;
for(i = 0; i < size; i++) {
if(arr[i] == element) {
return i;
}
}
return -1;
}
void deleteElement(int arr[], int *size, int position) {
int i;
if((position < 0) || (position >= *size)) {
printf(—Error: Invalid position.\n\n);
return;
}
for(i = position; i < *size - 1; i++) {
arr[i] = arr[i + 1];
}
(*size) --;
printf(—Element deleted successfully.\n\n);
}
void displayArray(int arr[], int size) {
int i;
if(size == 0) {
printf(—Array is empty.\n\n);
return;
}
printf(—Array elements: \n);
for(i = 0; i < size; i++) {
printf(—%d\n, arr[i]);
}
printf(—\n\n);
}

```

Output:

Array Operations Menu:

1. Create Array
2. Insert Element
3. Search Element
4. Delete Element
5. Display Array
6. Exit

Enter your choice: 1

Enter the number of elements: 5

Enter 5 elements: 1 2 3 4 5

Array created successfully.

Array Operations Menu:

1. Create Array
2. Insert Element
3. Search Element
4. Delete Element
5. Display Array
6. Exit

Enter your choice: 2

Enter element to insert: 10

Enter position to insert (0 to 5): 2

Element inserted successfully.

Array Operations Menu:

1. Create Array
2. Insert Element
3. Search Element
4. Delete Element
5. Display Array
6. Exit

Enter your choice: 5

Array elements: 1 2 10 3 4 5

Array Operations Menu:

1. Create Array
2. Insert Element
3. Search Element
4. Delete Element
5. Display Array
6. Exit

Enter your choice: 3

Enter the element to search: 10

Element found at position: 2

Array Operations Menu:

1. Create Array
2. Insert Element
3. Search Element
4. Delete Element
5. Display Array
6. Exit

Enter your choice: 4

Enter position to delete (0 to 5): 2

Element deleted successfully.

Array Operations Menu:

1. Create Array
2. Insert Element
3. Search Element
4. Delete Element
5. Display Array
6. Exit

Enter your choice: 5

Array elements: 1 2 3 4 5